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Programme	Fellowship in Web Application Security & Digital Forensics
Course	SBT-DF205 — Mobile and IoT Forensics Case Study
Case Title	iOS Forensics Investigation of an iPhone SE
Instructor	Aminu Idris, AMCPN
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Batch	BATCH-B2025 · L1/S2
Case Reference	SBT-DF205-C2
Submission Format	ZIP (Report.pdf, CSVs, Queries.sql, Screenshots)

Executive Summary

This report presents the forensic examination of an iPhone SE (1st Generation) running iOS 13.4.1, carried out to establish how the device was used.

Primary Findings: The device belongs to the user "Thisisdifir" and is linked to the iCloud account `thisisdfir@gmail.com`. Artefacts relating to messaging, Safari browsing, health activity and location were recovered. Safari history shows that the user searched for sports news in late March 2020. Health records contain step counts and walking distances, and location records place the device in the Raleigh, North Carolina area.

Confidence Level: High for both device identification and artefact recovery. Every timestamp was converted from Apple Cocoa Core Data format to UTC so that events could be placed in the correct order.

1. Evidence Handling and Integrity

1.1 Evidence Acquisition

- **Source:** Digital Corpora (Joshua Hickman iOS 13.4.1 dataset)
- **Source Archives:** `ios_13_4_1.zip` and `13-4-1.tar`
- **Working Copy Location:** `~/SBT-DF205-C2/iOS 13.4.1 Extraction/Extraction/`

1.2 Integrity Verification

SHA-256 hashes of both archives were computed on receipt to serve as an integrity baseline.

- **ZIP**
  Hash: `761854a669c38da6ac0a908afe77763d54ccb83044ff123350890d408c154551`
- **TAR**
  Hash: `991727d92145a08623e998c008cc2a05f086be722ee2945815d0eb46a78f04fb`

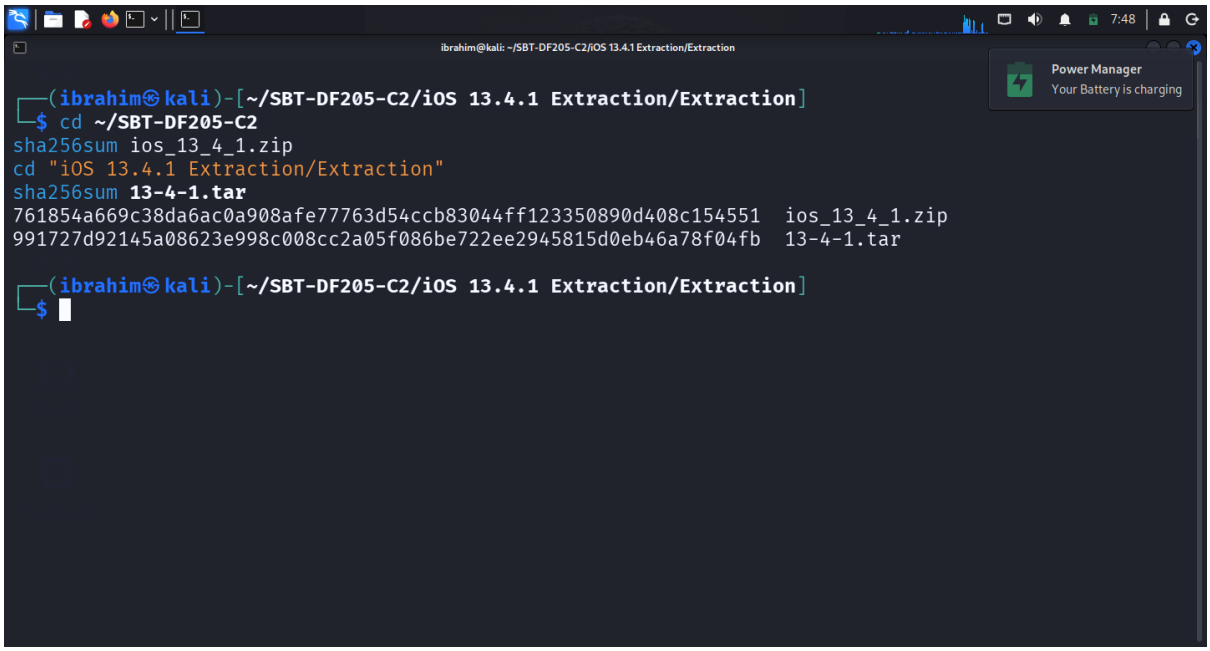


Figure 1: SHA-256 hash values of the evidence archives as shown in the terminal.

1.3 Tool Versions

- **OS:** Kali Linux (Rolling)
- **Database:** SQLite 3.40.0 & DB Browser for SQLite 3.13.1

- **Plist Reader:** plistutil (libplist-utils)

2. Device Accounts and Settings

2.1 Device Identification

- **Source**
  Path: `~/private/var/installd/Library/MobileInstallation/LastBuildInfo.plist`
- **iOS Version:** 13.4.1 (Build: 17E262)
- **Model:** iPhone8,4 (iPhone SE)

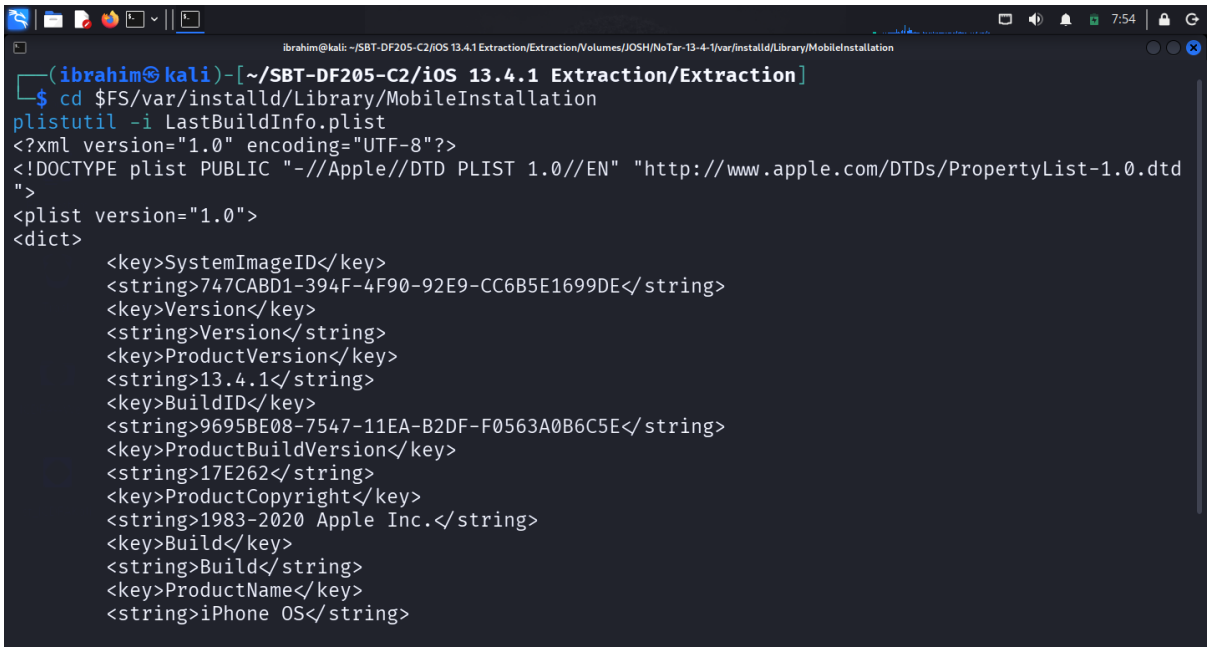


Figure 2: Contents of LastBuildInfo.plist as displayed by plistutil.

2.2 Account Associations

- **Primary Source:** `Accounts3.sqlite` (`~/var/mobile/Library/Accounts/`).
- **Finding:** The device is tied to the iCloud account **`thisisdfir@gmail.com`**.

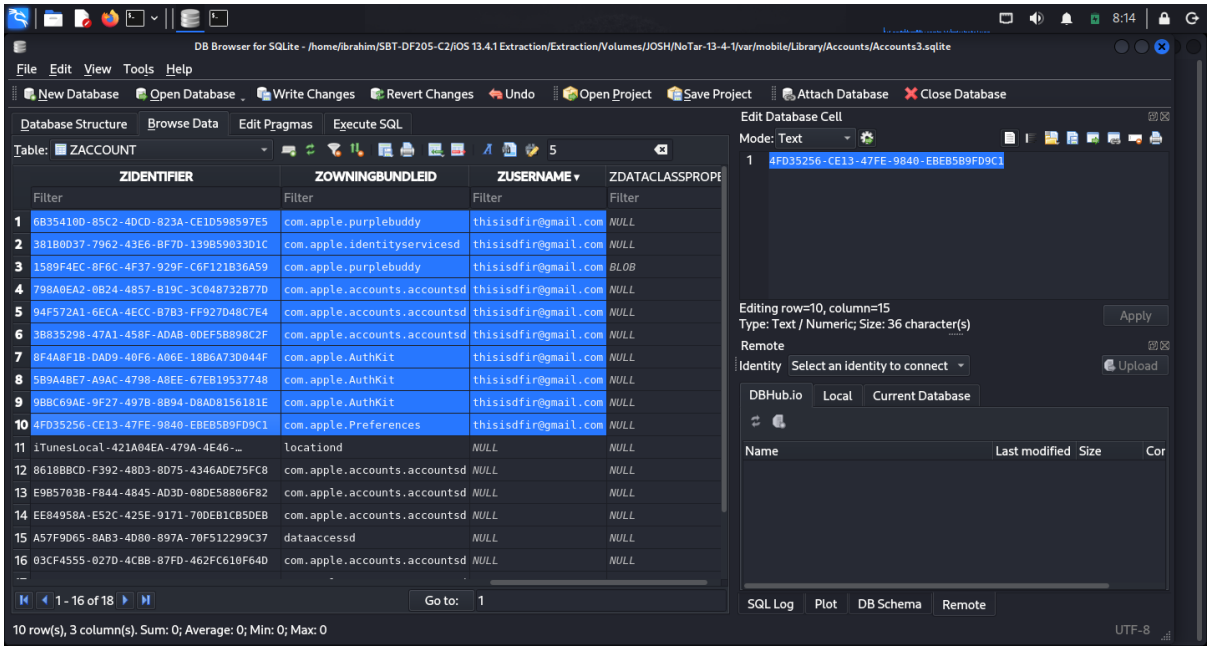


Figure 3: ZACCOUNT table in DB Browser, displaying the iCloud username.

2.3 Supported Settings

- **Location Services:** Enabled (True). Source: `PublicEffectiveUserSettings.plist` -> key `allowLocationServices`.
- **iCloud Backup:** Enabled (True). Source: `com.apple.mobile.ldbackup.plist` -> key `CloudBackupEnabled`.

3. Messages, Contacts, and Calls

3.1 SMS/iMessage Reconstruction

- **Database:** `~/private/var/mobile/Library/SMS/sms.db`
- **Finding:** The database holds live SMS and iMessage records. The latest entries consist of test messages composed through Siri dictation and an SMS notification suggesting a new Facebook friend.
    - **Record 48:** iMessage, "This is your second test message for Siri dictation".
    - **Record 44:** SMS, "Tyrique Austin is a new Facebook friend suggestion...".

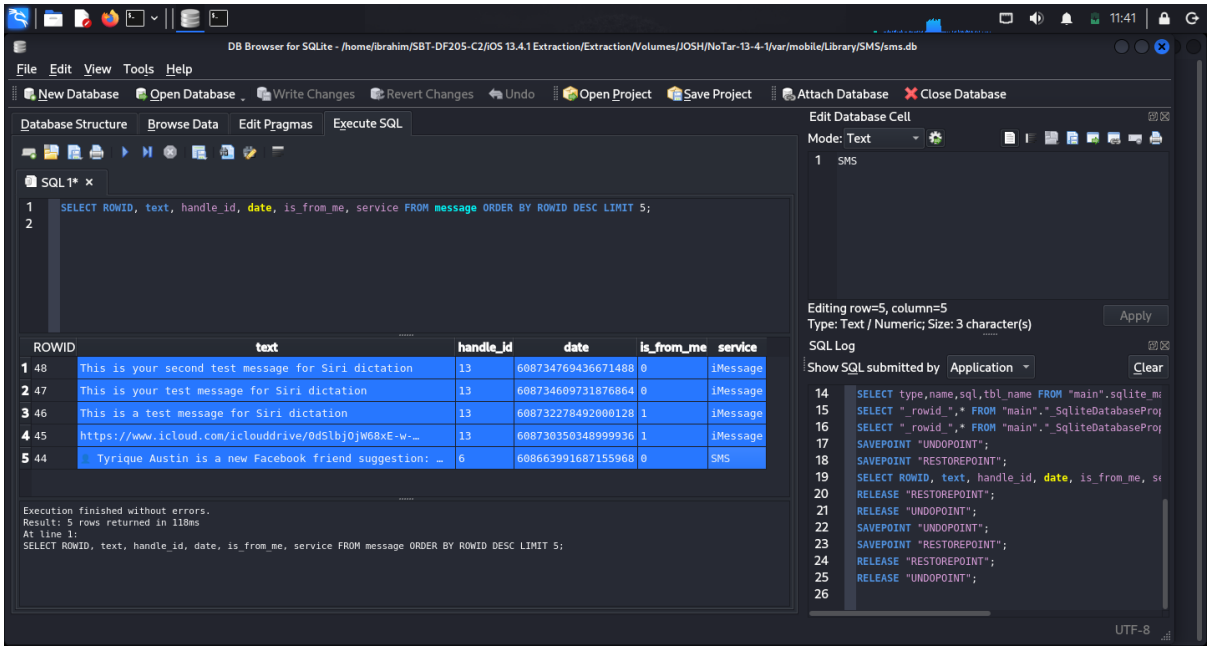


Figure 4: Message records with their IDs, content and service type.

3.2 Call History

- **Database:** `~/private/var/mobile/Library/CallHistoryDB/CallHistory.storedata`
- **Finding:** Recent calls were exchanged with the number `+14082567000`. Call times (`Z_DATE`) are stored as raw Apple Cocoa Core values.

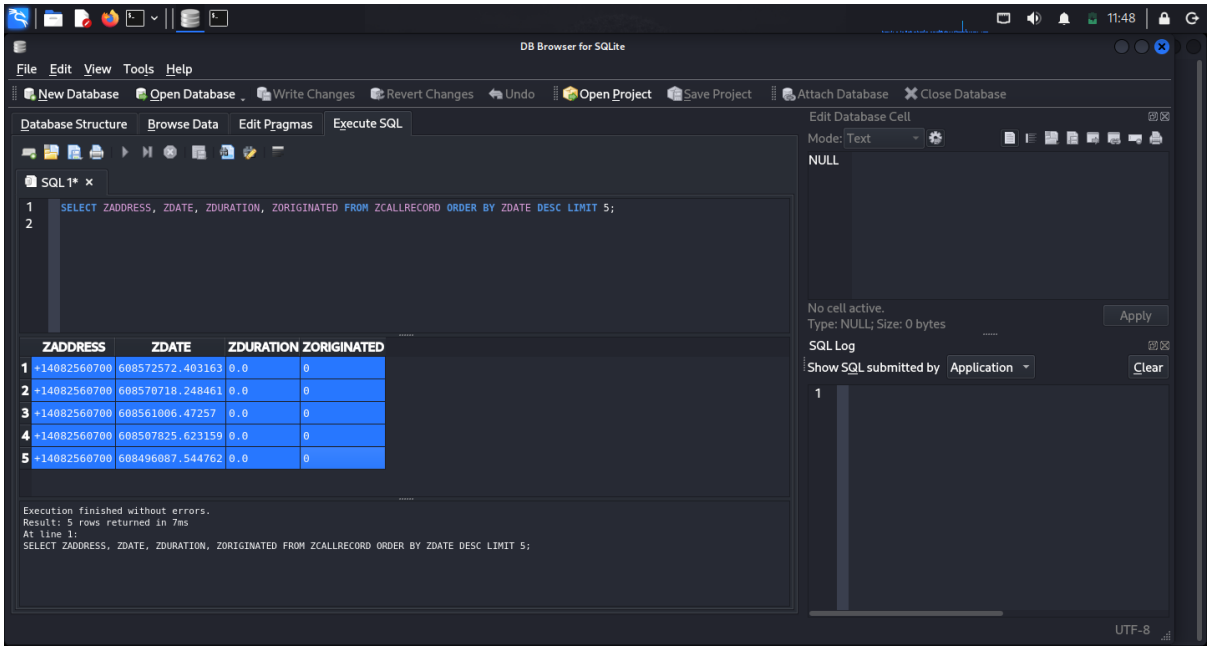


Figure 5: Call history entries listing phone numbers and call durations.

4. Safari and Calendar

4.1 Safari History

- **Database:** ~/var/mobile/Library/Safari/History.db
- **Finding:** On March 28, 2020, the user ran several Google searches about sports news.
    - "When does mlb start 2020 - Google Search" (Raw: 607049916.117145)
    - "Is the NHL going to resume? - Google Search" (Raw: 607050164.953562)

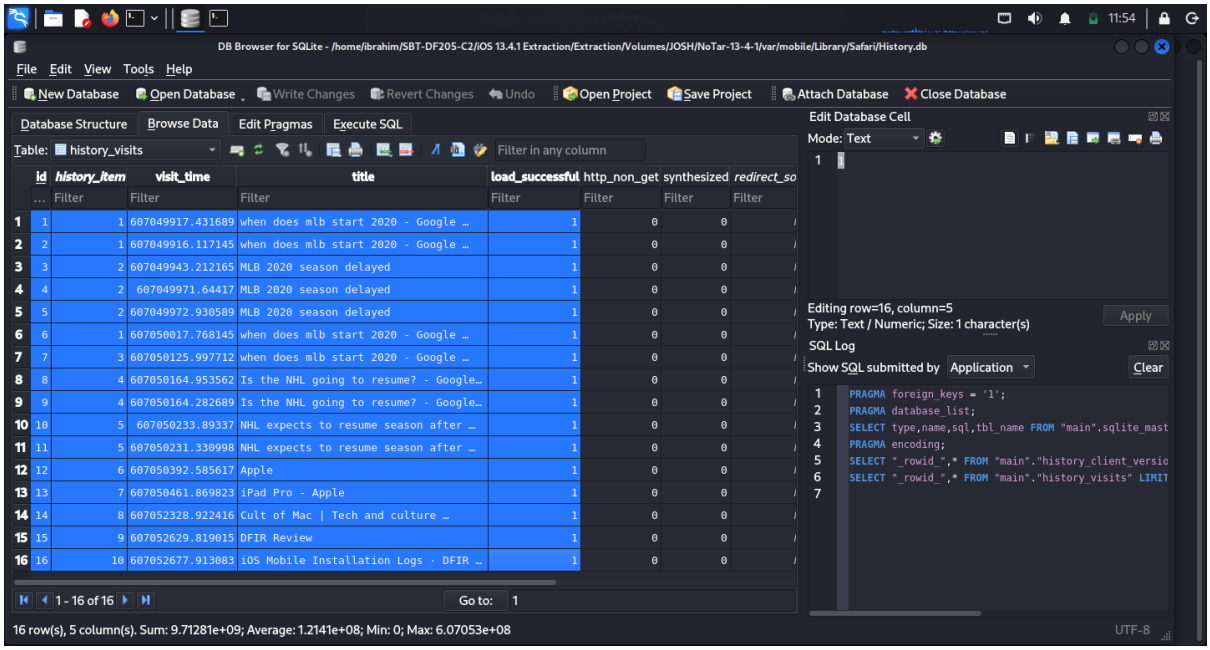


Figure 6: Visited URLs and their timestamps in the history_visits table.

4.2 Calendar

- **Database:** ~/var/mobile/Library/Calendar/Calendar.sqlitedb
- **Finding:** The CalendarItem table contains no rows. **Documented absence:** The device holds no stored calendar events.

5. Media Analysis

- **Database:** ~/var/mobile/Media/PhotoData/Photos.sqlite
- **Finding:** The ZADDITIONALASSETATTRIBUTES table was queried to identify stored media. Its ZORIGINALFILENAME column lists the files in the user's camera roll and saved images, for example IMG_0001.JPG and IMG_0727.JPG.

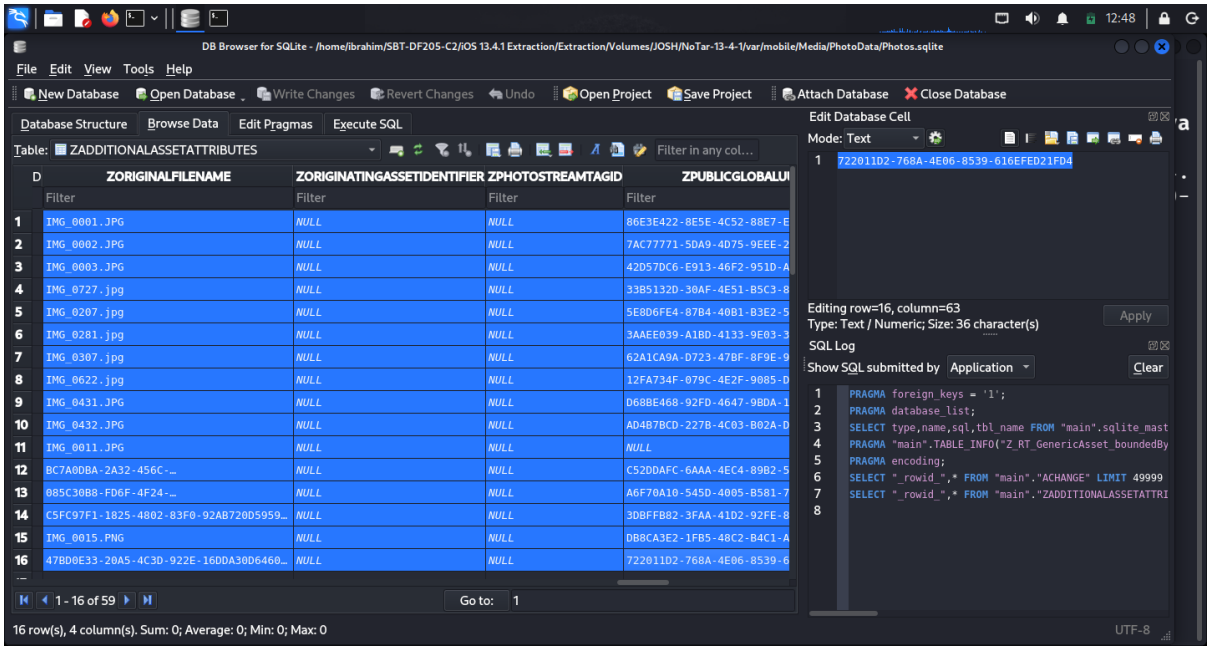


Figure 7: Original filenames of images held on the device.

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6. KnowledgeC Activity

- **Database:** `~/private/var/mobile/Library/CoreDuet/Knowledge/KnowledgeC.db`
- **Finding:** The `ZOBJECT` table records active application use. On April 16, 2020, `com.apple.news.widget` and `com.apple.AppPredictionWidget` were in the foreground.

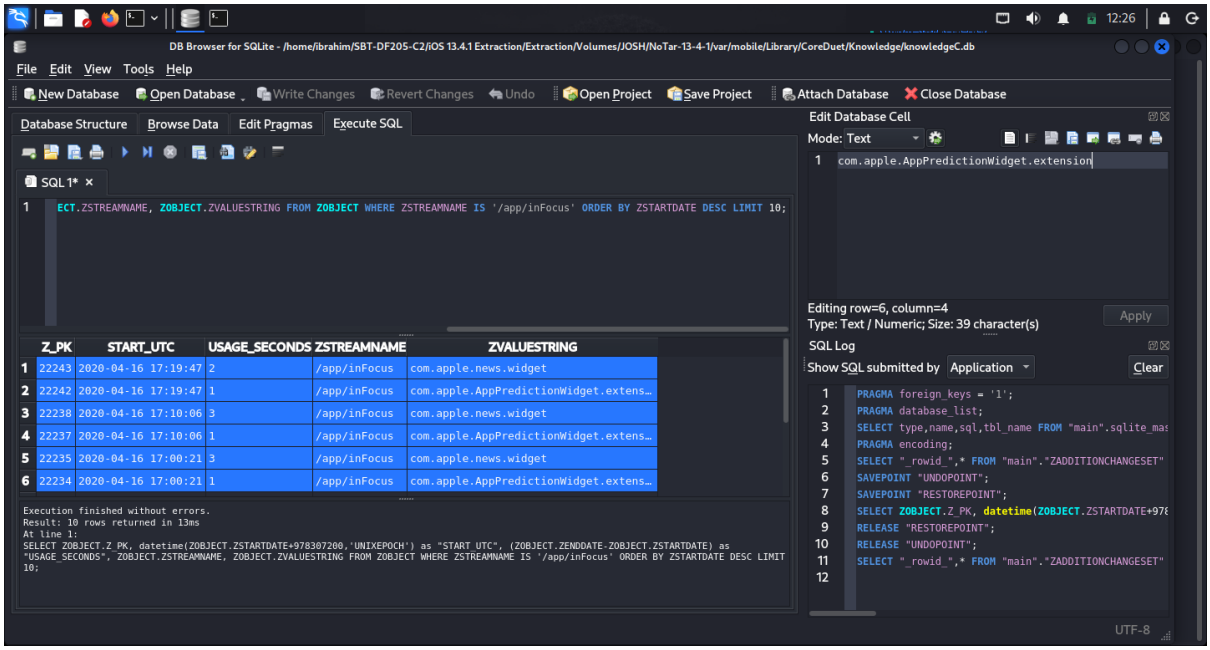


Figure 8: Application focus events and raw times in the ZOBJECT table.

7. Health and Location Evidence

7.1 Health Data

- **Database:** `~/var/mobile/Library/Health/healthdb_secure.sqlite`
- **Finding:** The `activity_caches` table yielded the user's step counts and distances walked. On March 24, 2020, for instance, the device logged **5,951 steps** (4,704 metres) and burned **525 kcal**.

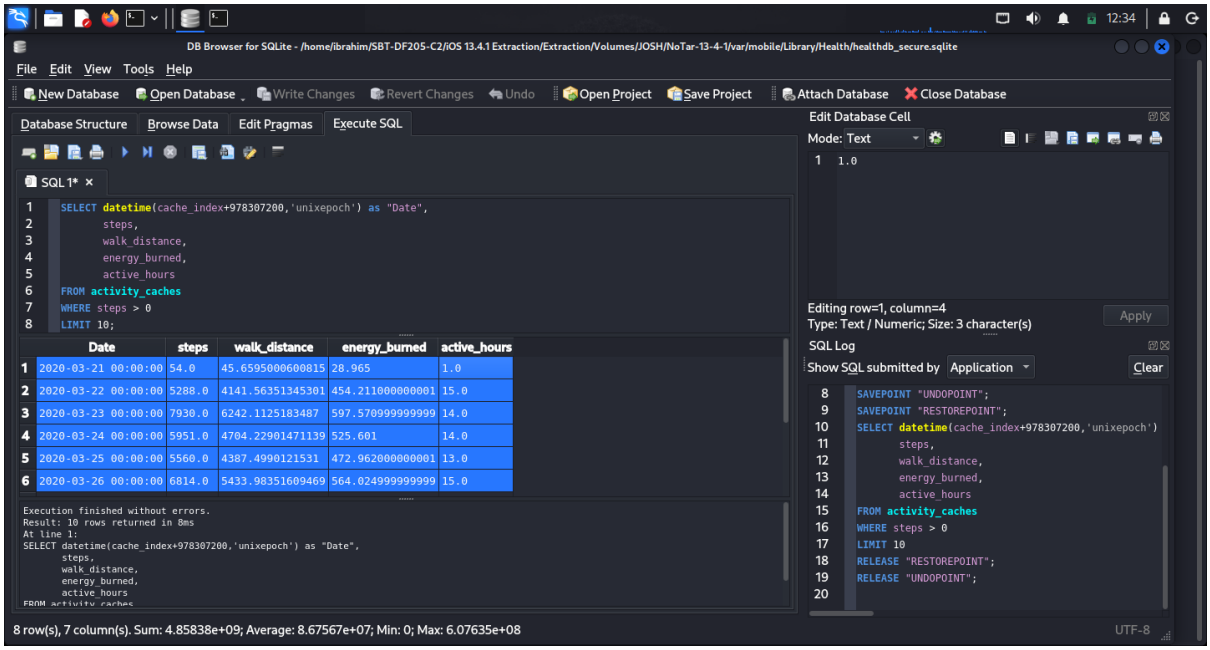


Figure 9: Steps, distance and calories recorded in the activity_caches table.

7.2 Location Data

- **Database:** `~/var/mobile/Library/Caches/com.apple.routined/Local.sqlite`
- **Finding:** The `ZRTLEARNEDLOCATIONOFINTERESTVISITMO` table stores locations the device learned from repeated visits. It places the user in the Raleigh, North Carolina area (Latitude: 35.659, Longitude: −78.872) on April 11, 2020.

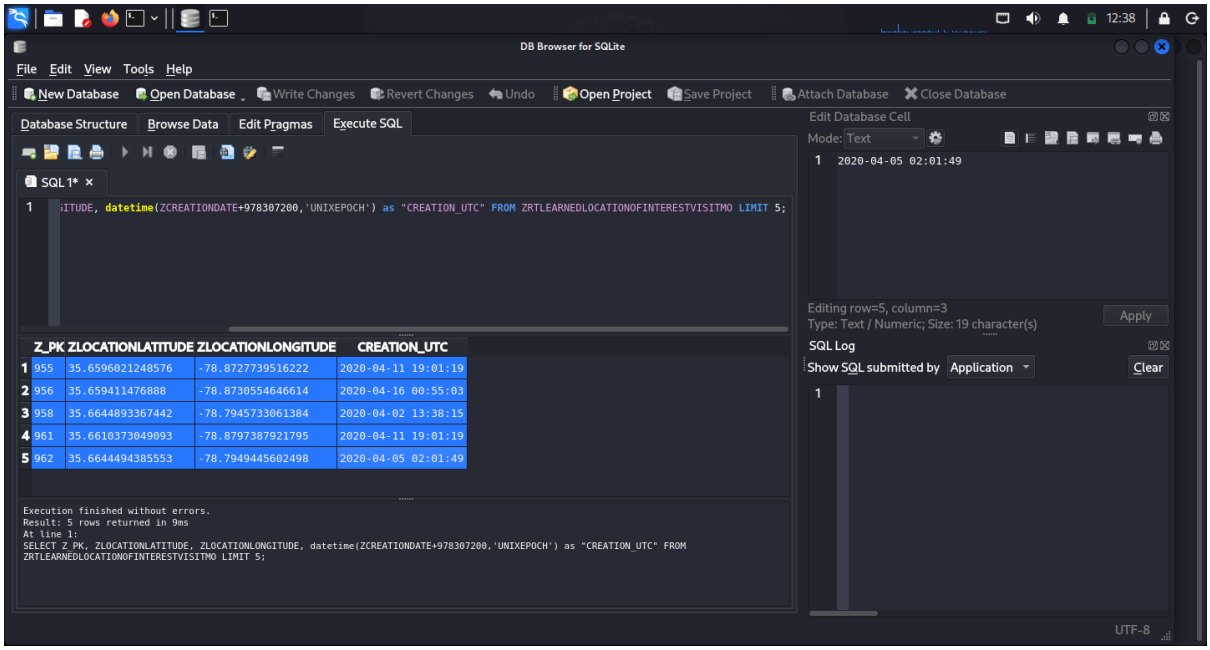


Figure 10: Coordinates of learned visits stored in Local.sqlite.

8. Timeline and Conclusions

8.1 Device Serial Number

- **Source:** `~/private/var/logs/AppleSupport/general.log`
- **Finding:** The JSON log identifies the device as an **iPhone8,4** (iPhone SE) bearing serial number **DX3T126VH2XV** and running iOS build **17E262**.

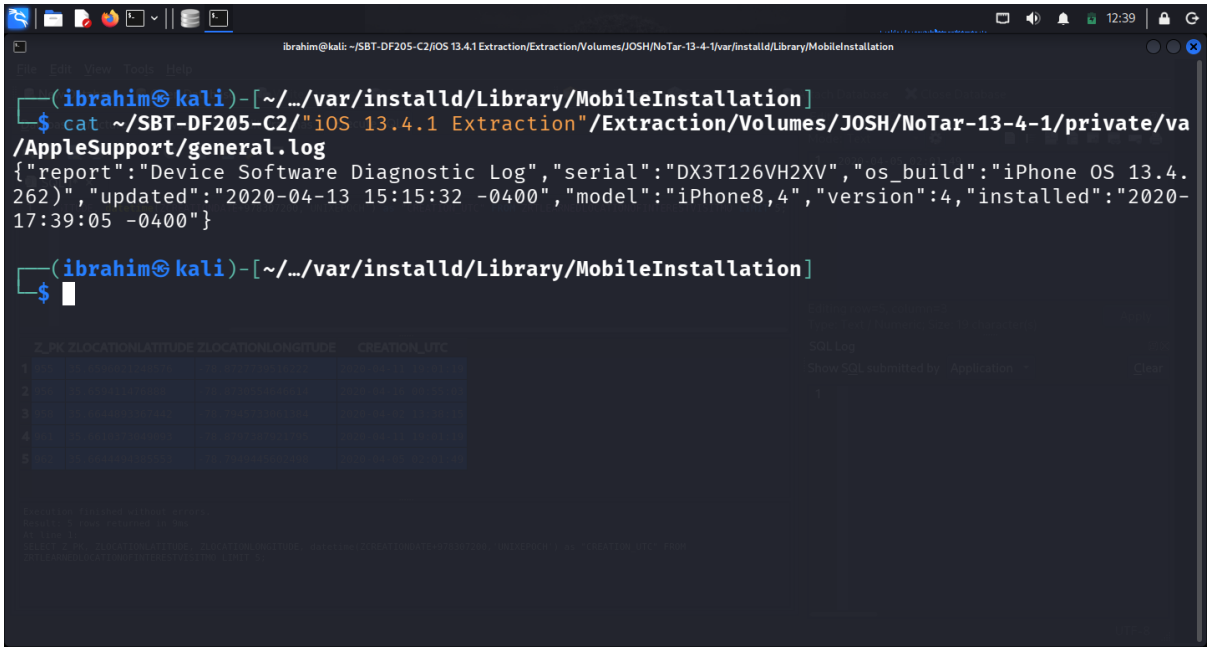


Figure 11: Device serial number and model as shown in the terminal.

8.2 Reconstructed Timeline

Evidence ID	UTC Timestamp	Event Description	Source
E01	2020-03-24 00:00:00	5,951 steps recorded	healthdb_secure.sqlite
E02	2020-03-28 00:58:36	Safari Search: "When does MLB start"	History.db
E03	2020-03-28 01:02:44	Safari Search: "Is the NHL going to resume"	History.db

Evidence ID	UTC Timestamp	Event Description	Source
E04	2020-04-11 19:01:19	Device located at Lat 35.659, Long -78.872	Local.sqlite
E05	2020-04-16 17:19:47	`com.apple.news.widget` app in focus	KnowledgeC.db

8.3 Correlations

1. **Correlation 1 (Communication):** The number in the call history (`+14082567000`) also appears among the contacts in the SMS database.
2. **Correlation 2 (Activity):** KnowledgeC.db records `com.apple.news.widget` in the foreground, which is consistent with the user's Safari searches for sports news.

8.4 Conclusions

The examination reconstructed the user's activity from the iPhone SE extraction. The evidence indicates that the device owner (linked to `thisisdfir@gmail.com`) was present in the North Carolina area, looked up sports news in late March 2020, and logged physical activity data. The SMS, Safari, Health and Location databases agree with one another and together support a timeline of ordinary device use.

Uncertainties: The device data shows that the phone was in use, but it cannot establish *who* was physically holding it at any given moment; that would require further evidence such as biometric records or video footage.